Welcome



BSF II Module

L1 Functional Organization of The Human Body

Objectives

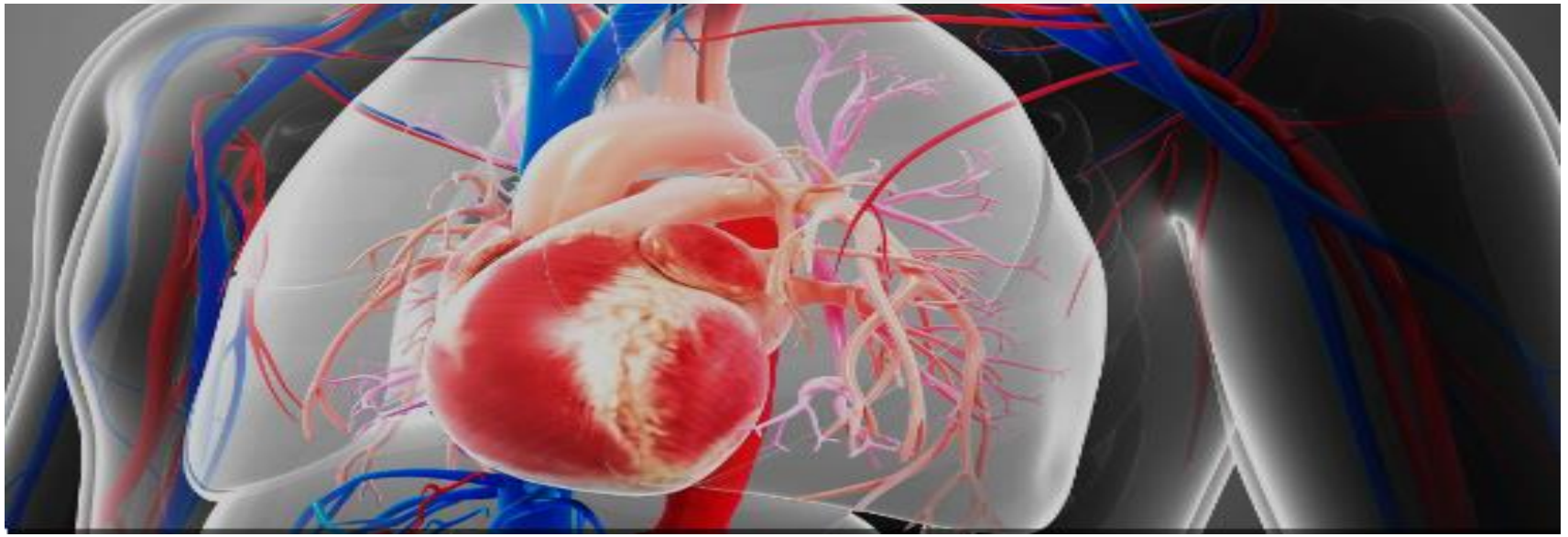
After studying this section, you should be able to:

- **Define:**
  - a. Physiology
  - b. Human physiology
  - c. Cell
- **Describe** the main functions of (functional organization) of human body systems.
- **Understand** the distribution of total body water and the clinical significance of each fluid compartment.

Functional Organization of the Human Body

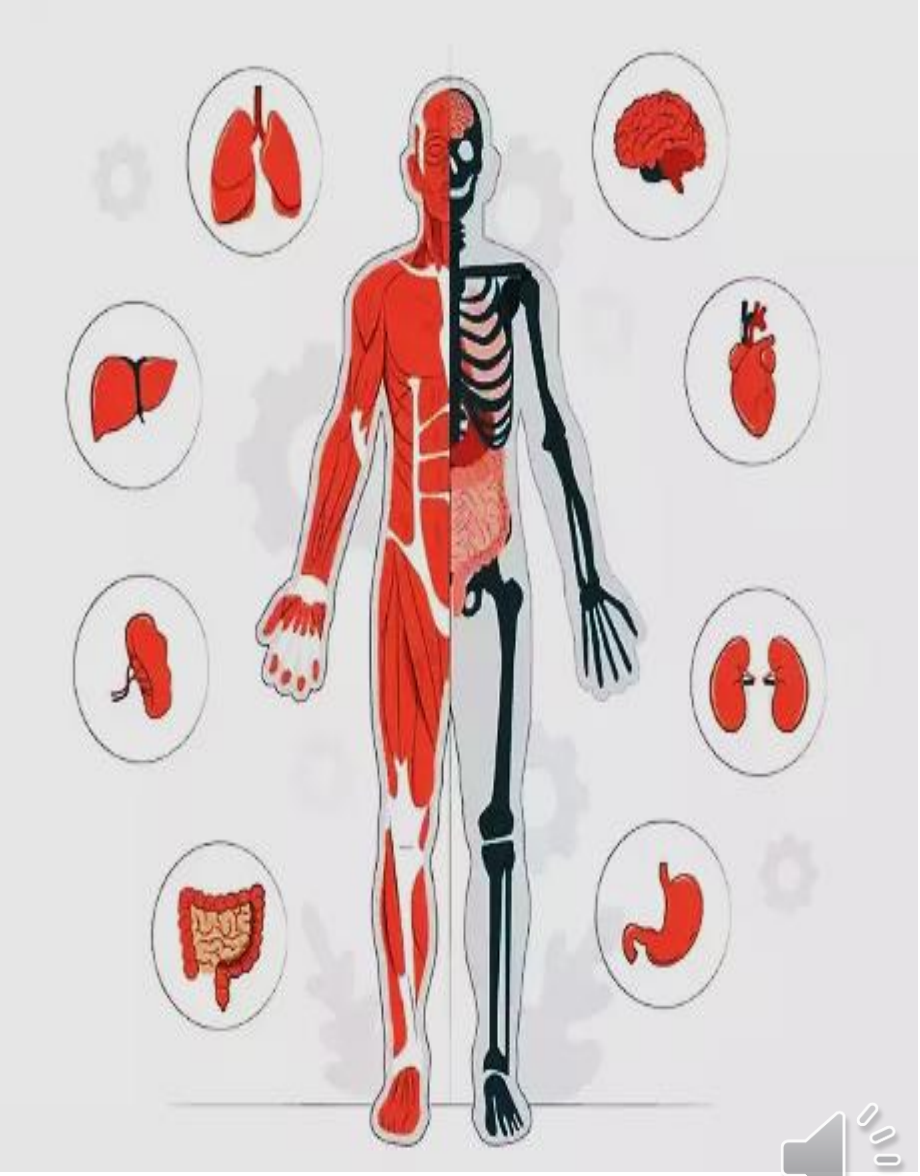
Outlines

- Definition of Human physiology
- Organization of the human body
- Composition of the human body
- Body water & its distribution



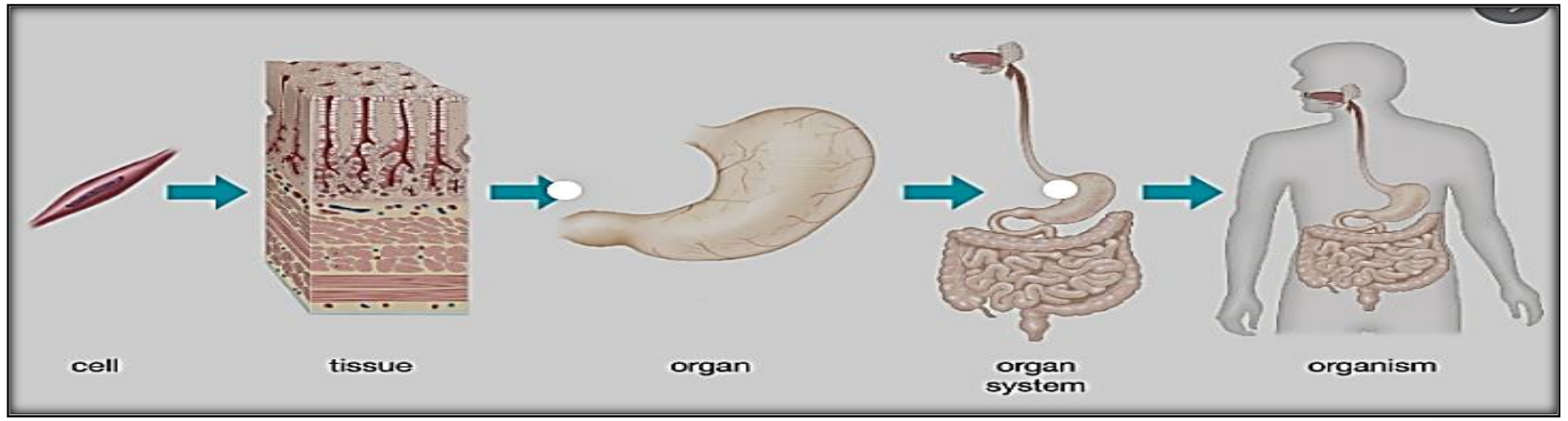
Definition of Human physiology

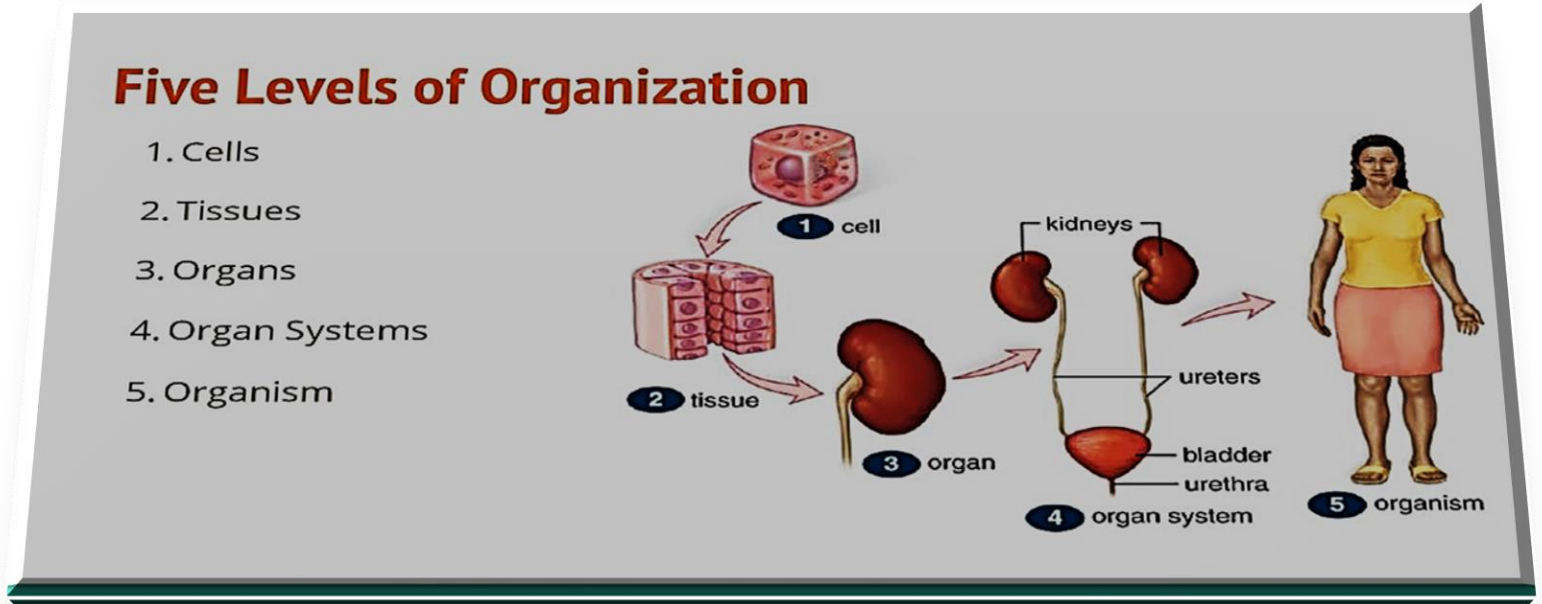
- The **science** concerned with studying **the normal functions** of the different parts of the human body and to explain **mechanisms** that keep the human being **alive**



Organization of The Human Body

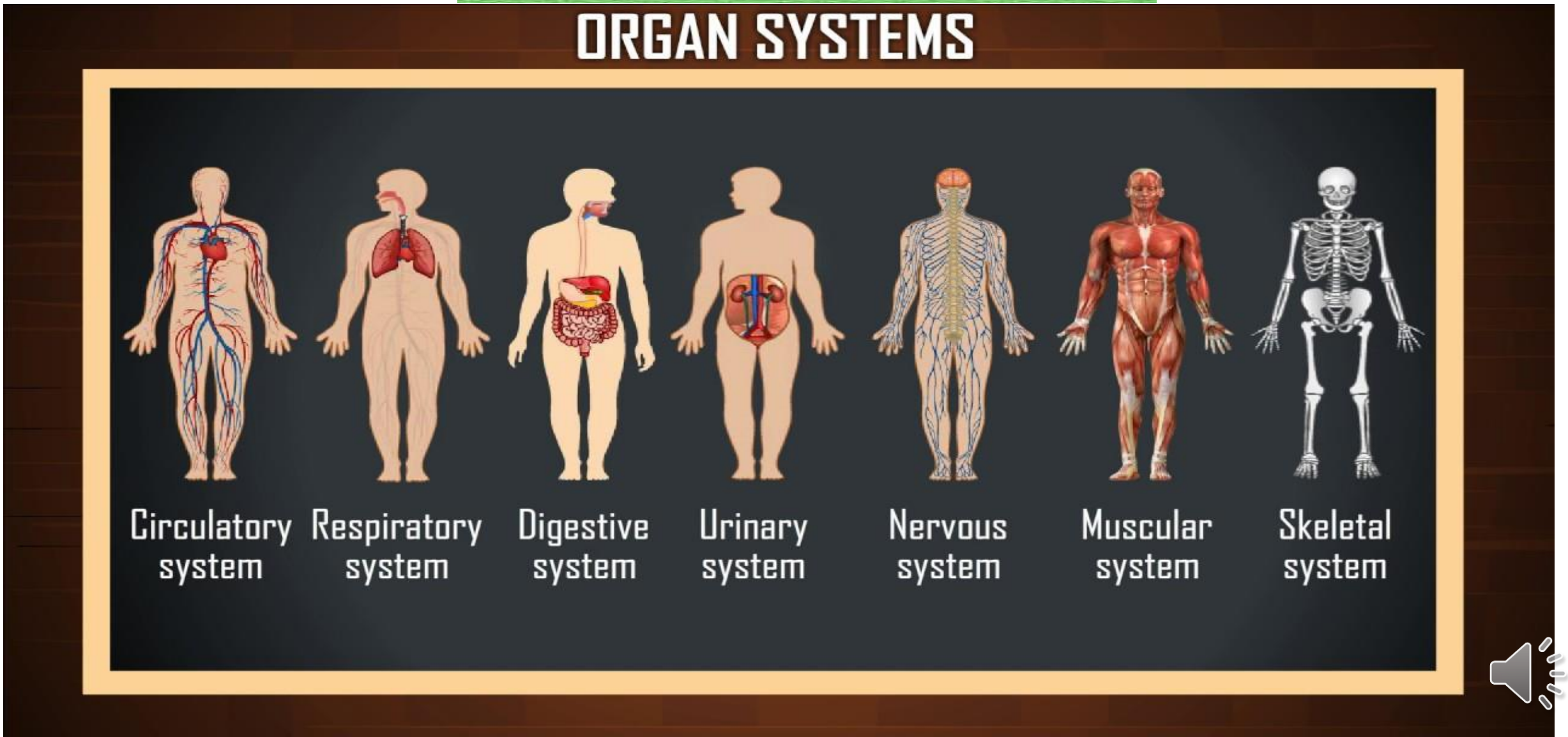
Cell → Tissue → Organs → System → Body





Systems

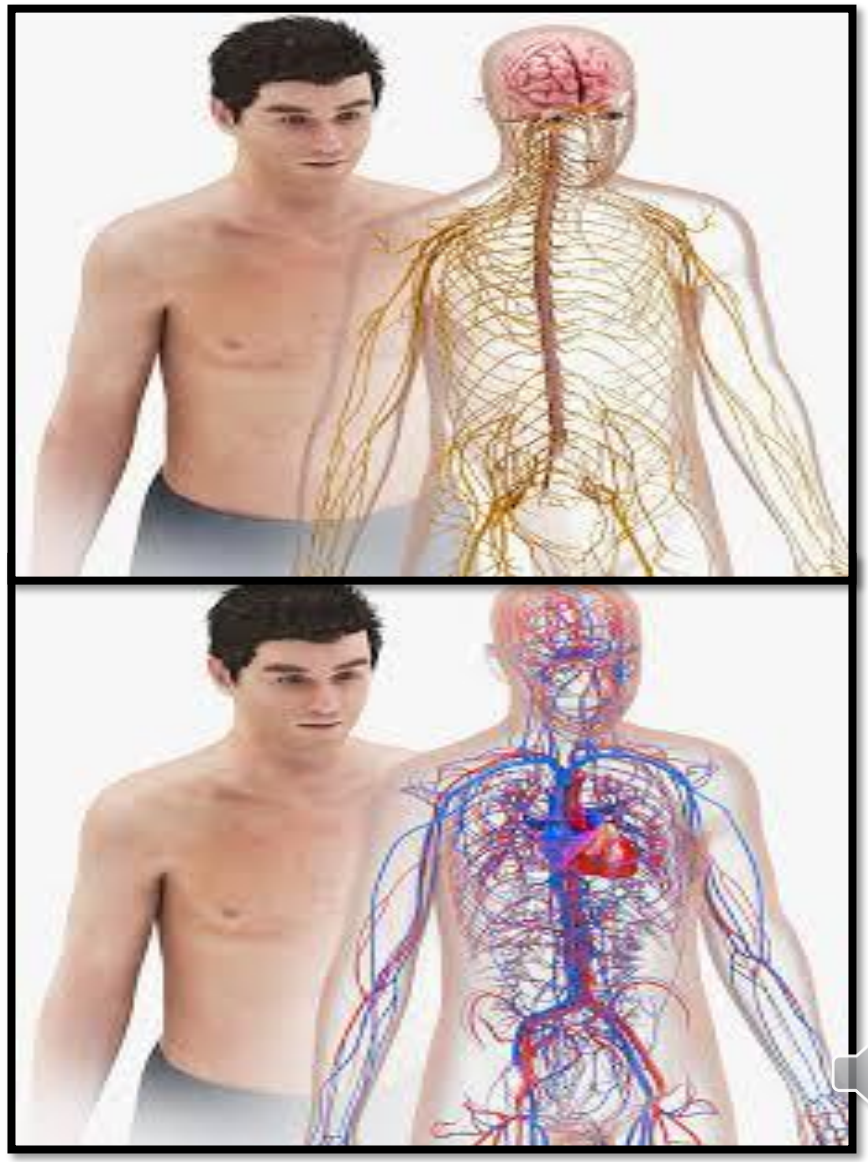
Body systems include:



Organs

Each of the body systems is formed of a group of organs

❖**Nervous system is formed of brain , spinal cord and nerves.**



❖**Cardiovascular system is formed of heart and blood vessels.**

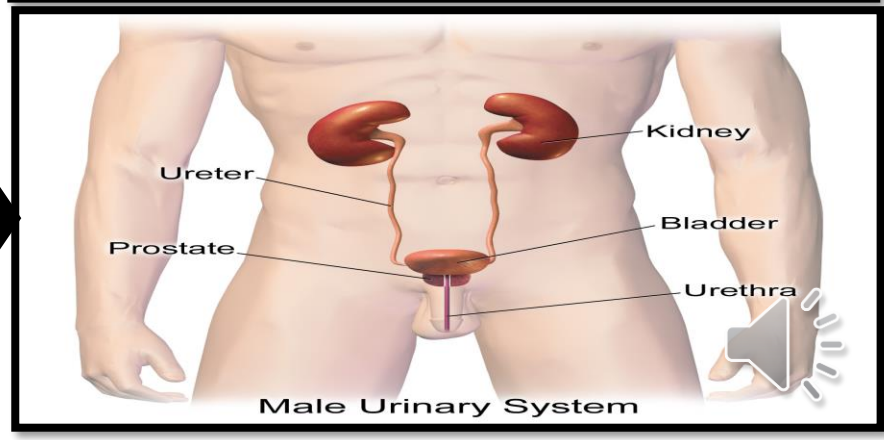
❖ **Respiratory system** is formed of nose, pharynx, larynx, trachea, bronchi, bronchioles and alveoli.



❖ **Gastrointestinal (digestive) system** is formed of mouth, pharynx, esophagus, stomach, small intestine and large intestine.



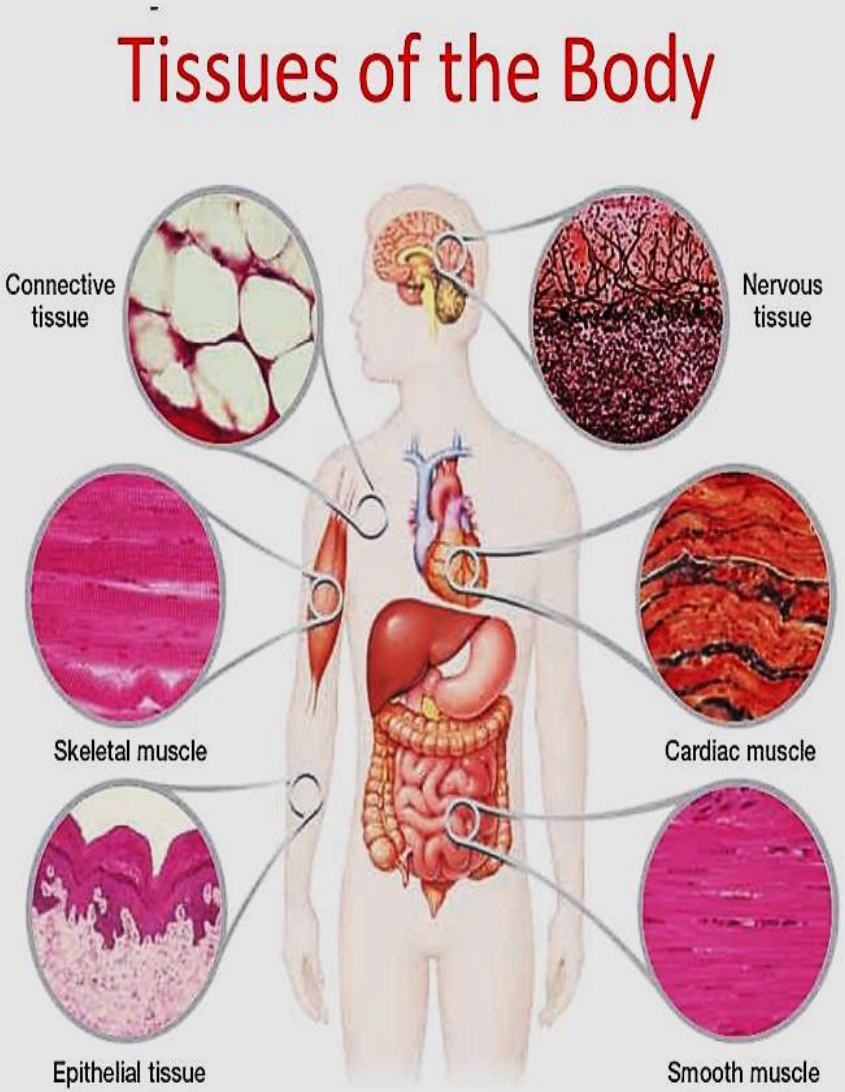
❖ **Urinary system** is formed of 2 kidneys, 2 ureters, urinary bladder and urethra.



Tissues

Each of the body organs is formed of a group of tissues. Tissues may be:

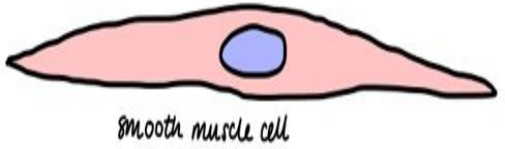
- **Nervous tissue.**
- **Muscular tissue.**
- **Fibrous tissue.**
- **Connective tissue.**
- **Epithelial tissue.**

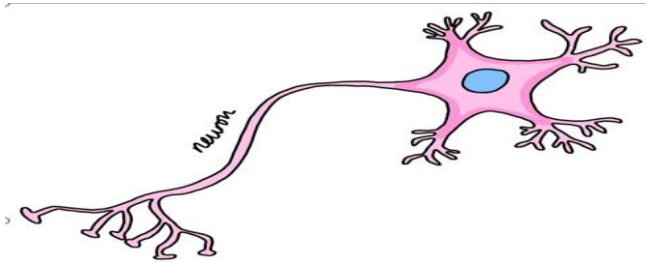


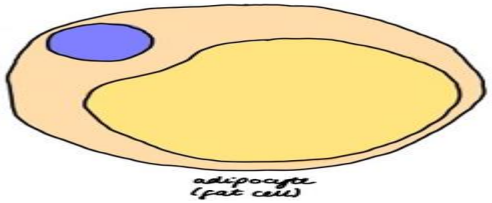
Cells

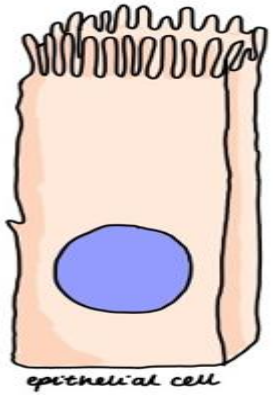
Each of the body tissues is formed of a group of cells. Cells may be:

- Nerve cells.
- Muscle cells.
- Connective tissue cells.
- Epithelial tissue cells.









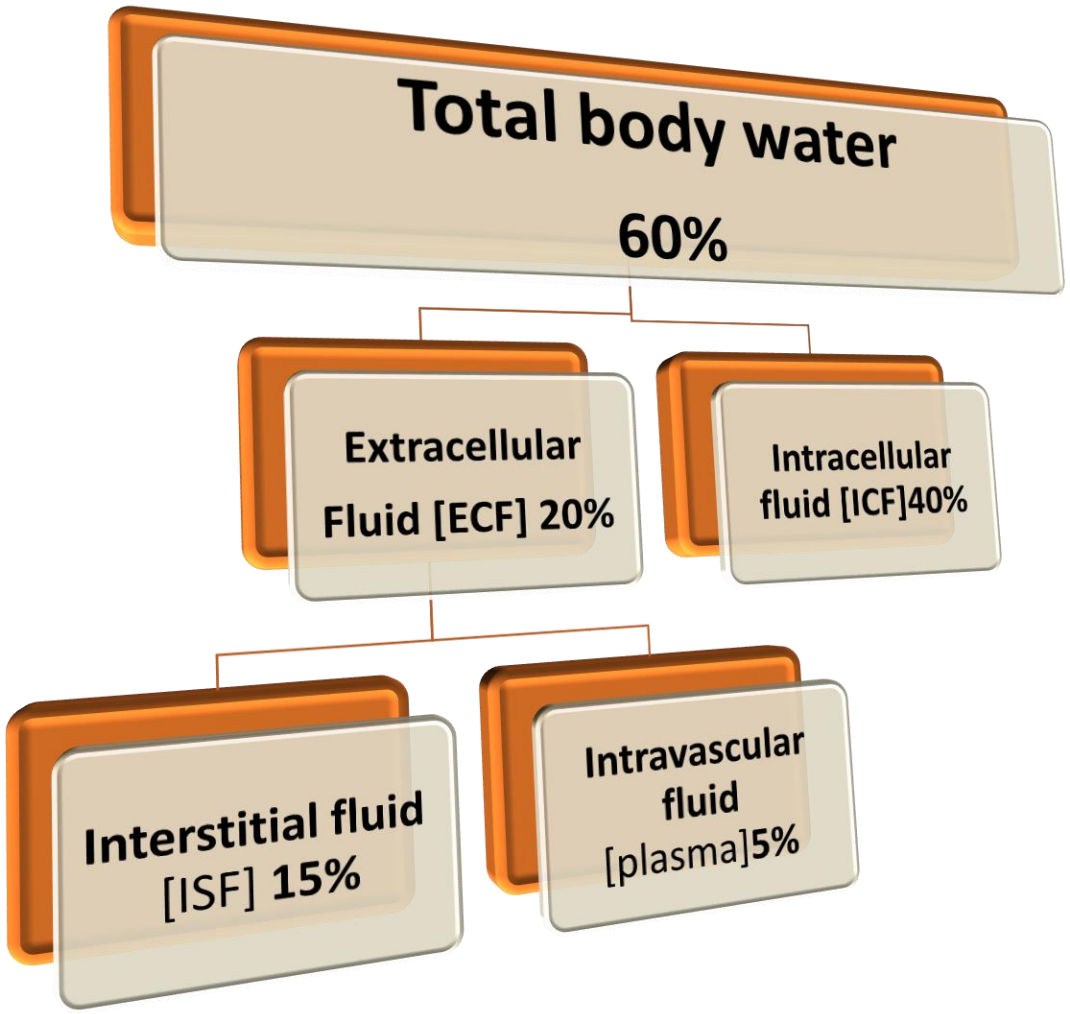
The cells is: the basic living unit of the human body. The human body contains about 100 trillion cells that work together for the maintenance of the living organism.

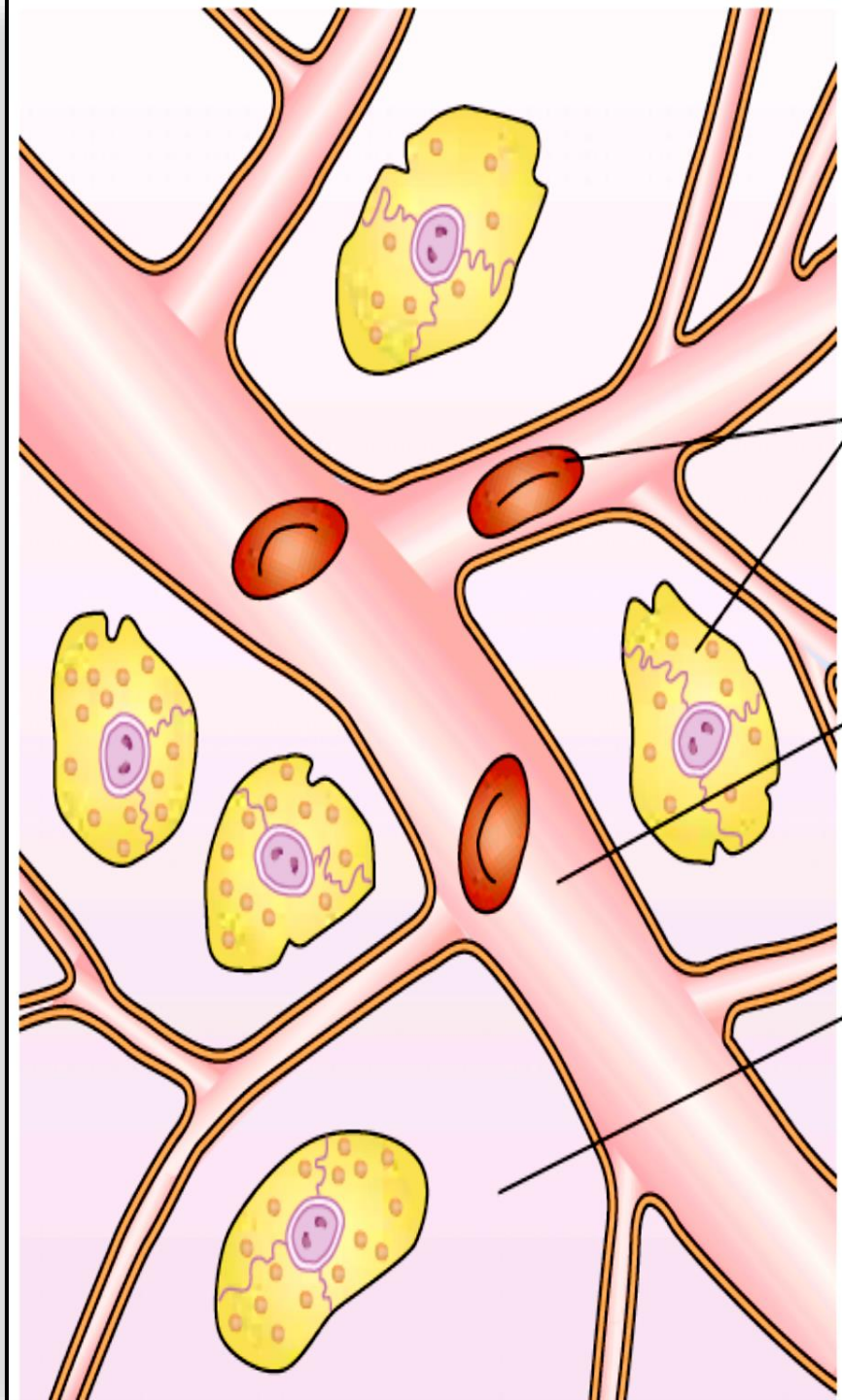
Composition of The Human Body

In average adult male, the body is composed of:

Water	60 %
Proteins	20 %
Fat	15 %
Minerals	5 %
Carbohydrates	Very little




Total body water
60%
Extracellular
Fluid [ECF] 20%
Intracellular
fluid [ICF]40%
Interstitial fluid
[ISF] 15%
Intravascular
fluid
[plasma]5%


Intracellular
water
Extracellular
(plasma) water
Extracellular
(interstitial) water

•Normal variation of body water:

- ✸ Body water is more in infants and children (about 75% of body weight).
- ✸ Body water is less in old age (about 45% of body weight).
- ✸ Body water is less in females (about 50% of body weight) due to high fat content as body fat is relatively free of water.
- ✸ Body water is less in obese person.

- **NB:** The increase of total body water in children is mainly in ECF.
- As ECF is easier to be lost in dehydration , thus dehydration develops more rapid and more sever in children than adults.



Distribution of ions in ICF and ECF:

	ECF	ICF
Cations		
Na+	140 mEq/L	14 mEq/L
K+	4 mEq/L	140 mEq/L
Ca++	2.4 mEq/L	0.0001 mEq/L
Mg++	1.2 mEq/L	58 mEq/L
Anions		
CL-	104 mEq/L	4 mEq/L
HCO3-	28 mEq/L	10 mEq/L
PO4---	4 mEq/L	75 mEq/L
Proteins	2 gm %	16 gm%

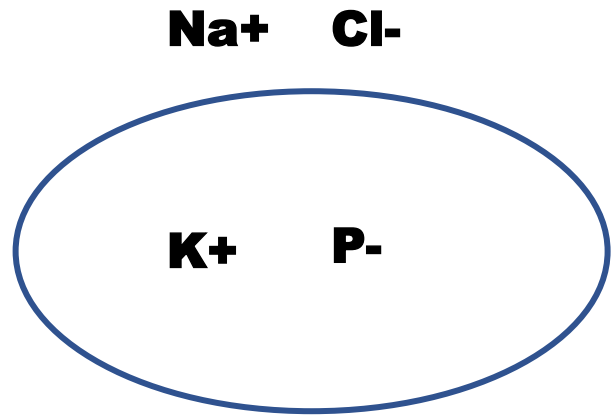


❖The main cation in ICF is K and in ECF is Na.

❖The main anion in ICF are phosphate then proteins.

❖The main anion in ECF are Cl- then Hco3.

❖Tonicity is the same between ICF and ECF.



❖The two components of ECF (ISF and plasma) are nearly similar in ionic composition except for protein which is higher in plasma.

Wrap-Up Questions

1-The main ICF cation is …… .

2-The main ECF cation is ……

3- The main ICF anion is ……

4- Regarding total body water, Which sentence is right?

A- It is less in male than female.
B- It is less in adults than older.
C- It is more in adults than children.
D- It is less in female than male.

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THANK
YOU!